

1(a)

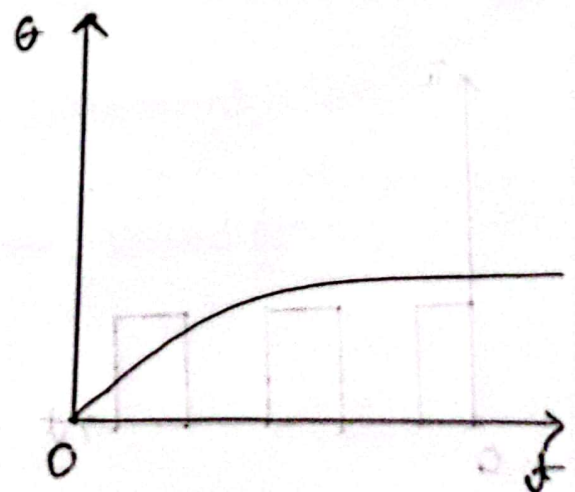
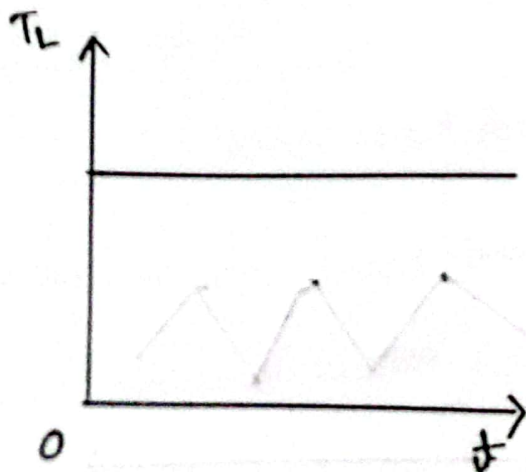
Faraday's second law: "The induced e.m.f in a coil is equal to the rate of change of flux linkage."

$$E = -N \frac{d\phi}{dt}$$

There are 8 kinds of motor duty class. In below some of them types of motor duty class been discussed:-

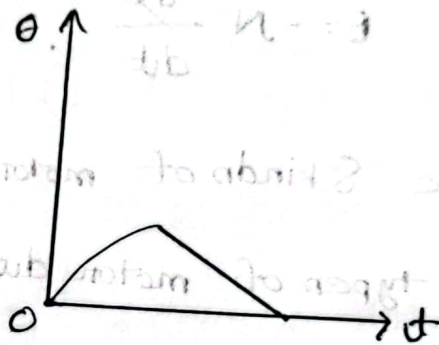
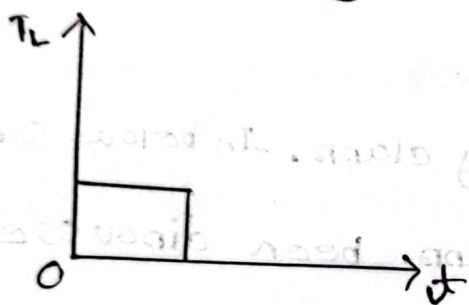
1. Continuous Duty Class

This duty denotes that, the motor is running long enough and the electric motor temperature reaches the steady state value. It's used in paper mill drivers, compressors etc.



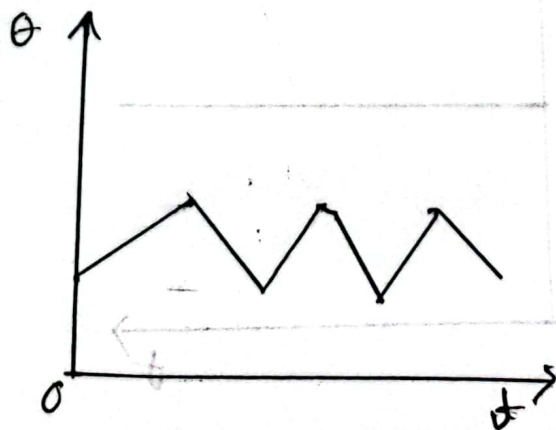
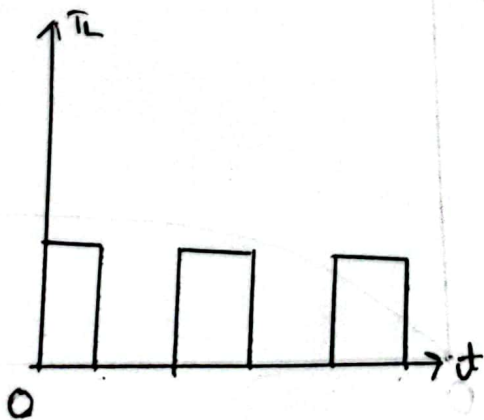
2. Short-time duty class

In these motors, the time of operation is very low and the heating time is much lower than the cooling time. So, the motor cools off to ambient temperature before operating again. It's used in crane drivers.



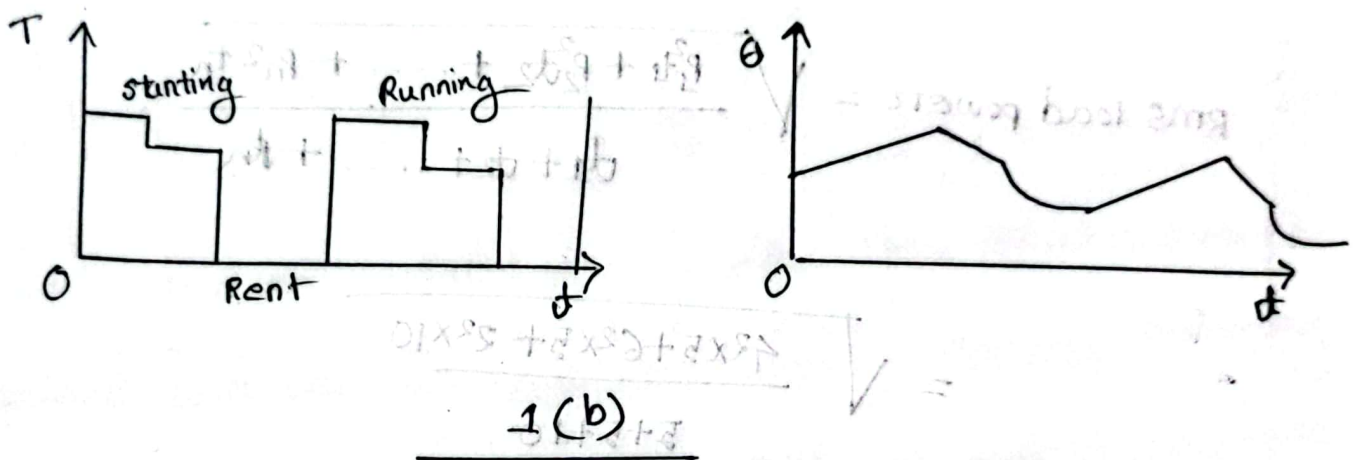
3. Intermittent Periodic Duty

Here motor operates for some time and there is a rest period. Both case, time is insufficient to raise the temperature to steady state value or cool it off to ambient temperature. Ex:- Drilling machine drivers



4. Intermittent Periodic Duty with Starting

There is a period of starting, which cannot be ignored and there is a heat loss at that time. After that there is running period and rest period which are not adequate to attain the steady state temperature.



Motor Sizing: The process of picking the correct motor

for a given load:

Sizing the motor:-

Given that,

Peak load power requirement = 6KW.

Peak load power must be less than Rated (motor power $\times$ Service factor)

$$\Rightarrow 6\text{KW} < P_{\text{motor}} \times 1.2$$

$$\Rightarrow P_{\text{motor}} > \frac{6}{1.2} = 5\text{KW}.$$

The RMS load power should ideally be between 75% and 100% of Rated motor power.

$$\text{RMS load power} = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + \dots + P_n^2 t_n}{t_1 + t_2 + \dots + t_n}}$$

$$= \sqrt{\frac{4^2 \times 5 + 6^2 \times 5 + 2^2 \times 10}{5 + 5 + 10}}$$

$$= 4.16\text{KW}.$$

So, if we choose $P_{\text{motor}} = 5\text{KW}$ then RMS load power

$$= \frac{4.16}{5} \times 100\% = 83\% \text{ of } P_{\text{motor}}, \text{ which is}$$

acceptable.

Working of DC Generator (operation) :-

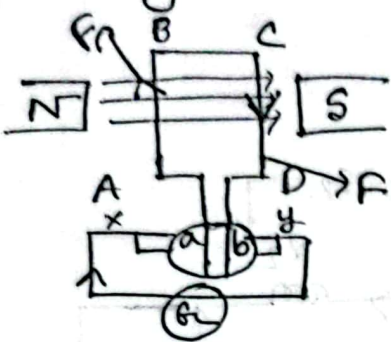


fig-1

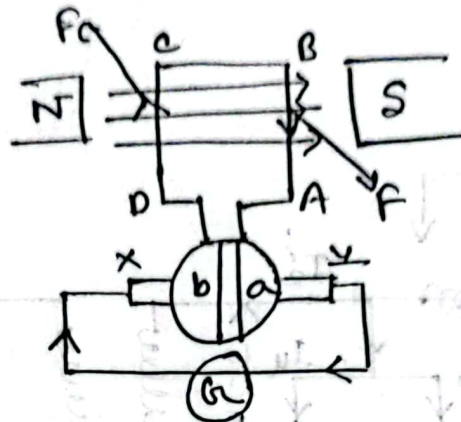


fig-2

Suppose the coil rotate clock wise, side AB of coil move upward and side CD move in downward. Applying Fleming's Right hand rule on side AB force is upward, magnetic field is left to right. So, current flows A to B.

Applying Right hand rule to CD side force downward, magnetic field is left to right. So, current flows from C to D.

Hence, current flows into Brush Y, move along galvanometer and finally enters X. Hence we can say current flows Y to X in the external circuit (fig-1)

After half rotation, side CD comes on left side and AB comes on right side.

Now, split rings b is connected to CD and a is connected to AB in fig-2. which keep the direction of current same. Due to commutator arrangement, the direction of current is not change.

Hence, current flows from brush Y, move along the galvanometer and finally enters X. Hence, we say current flows from Y to X in the external circuit.

Thus, after every half rotation, direction of current same. So, DC current is produced.

2(a)

The practical equivalent circuit of transformer with proper notation.

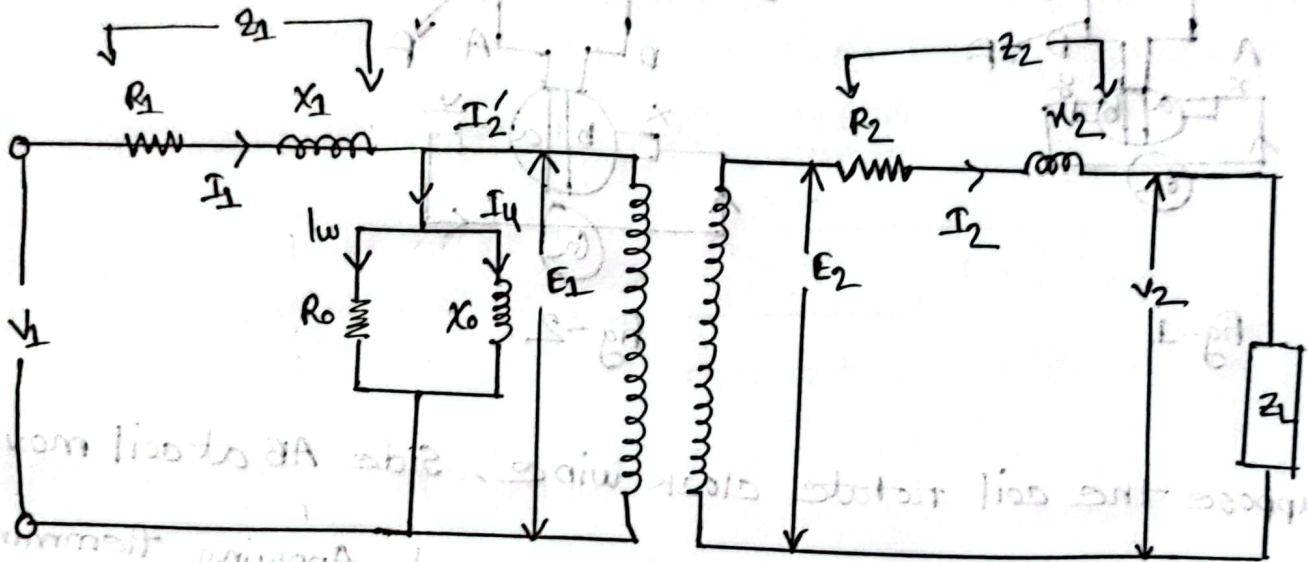


Fig: Ideal Transformer.

Where,

R_1 = Primary Winding Resistance

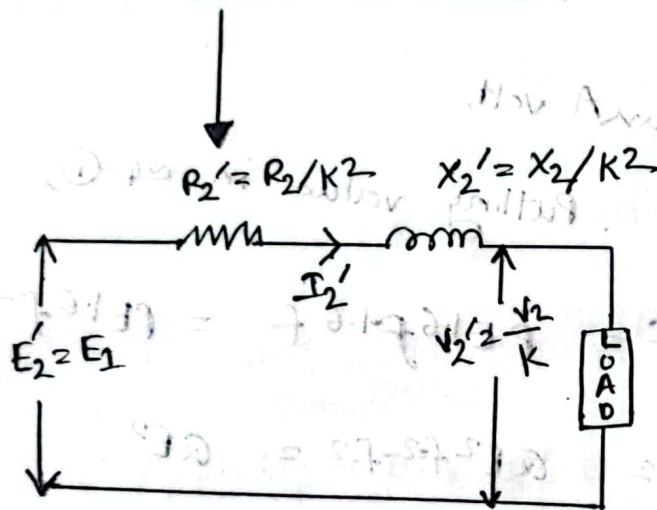
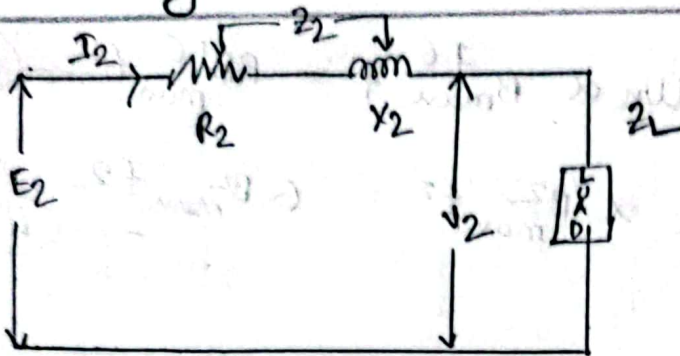
R_2 = Secondary " "

I_0 = No-load current

I_u = Magnetizing Component.

I_w = Working Component

Secondary Impedance referred to Primary

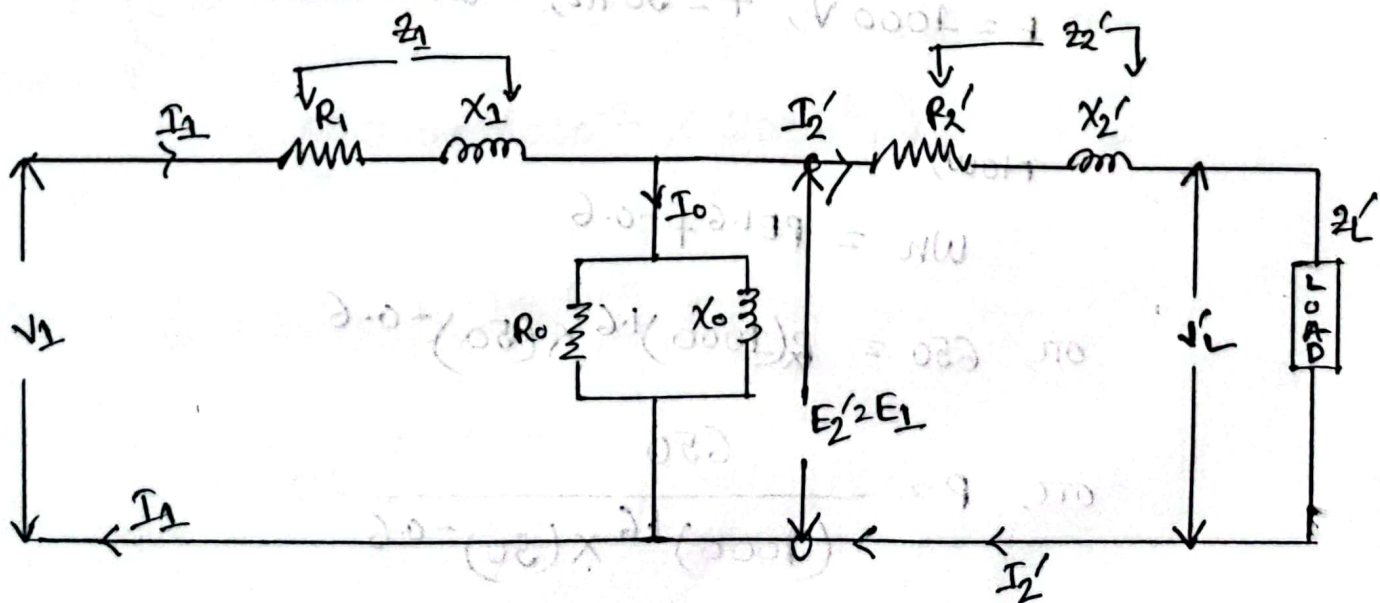


$$R_2' = R_2 / K^2$$

$$X_2' = X_2 / K^2$$

$$Z_L' = Z_L / K^2$$

Now, Equivalent Circuit - Referred to Primary



problem at 2(b)

Soln: Hysteresis loss $W_h \propto B_{\max}^{1.6} f = P B_{\max}^{1.6} f$ — (i)

Eddy current loss $W_e \propto B_{\max}^2 f^2 = Q B_{\max}^2 f^2$ — (ii)

from the relation,

$$E = 4.44 f N B_{\max} A \text{ volt,}$$

we get $B_{\max} \propto \frac{E}{f}$. Putting value in eq (i),

$$W_h = P \left(\frac{E}{f} \right)^{1.6} f = P E^{1.6} f^{-1.6} f = P E^{1.6} f^{-0.6}$$

$$W_e = Q \left(\frac{E}{f} \right)^2 f^2 = Q E^2 f^2 f^2 = Q E^2$$

Given,

$$E = 1000 \text{ V, } f = 50 \text{ Hz, } W_h = 650 \text{ W, } W_e = 350 \text{ W}$$

Now,

$$W_h = P E^{1.6} f^{-0.6}$$

$$\text{or, } 650 = P (1000)^{1.6} \times (50)^{-0.6}$$

$$\text{or, } P = \frac{650}{(1000)^{1.6} \times (50)^{-0.6}}$$

$$= 650 \times 1000^{1.6} \times 50^{0.6}$$

And

$$W_e = Q E^2$$

$$\text{m, } 350 = Q \times (1000)^2$$

$$\text{m, } Q = 350 \times (1000)^{-2}$$

$$\therefore P = 650 \times 1000^{-1.6} \times 50^{0.6} \text{ and } Q = 350 \times (1000)^{-2}$$

when voltage raised to 2000V and frequency

to 1000 Hz. The new core loss.

$$= (W_{h2} + W_{e2})$$

$$E_2 = 2000V$$

$$f_2 = 50Hz$$

$$= (P E_2^{1.6} f^{-0.6} + Q \times E_2^2)$$

$$= (650 \times 1000^{-1.6} \times 50^{0.6} \times 2000^{1.6} \times 50^{-0.6})$$

$$+ (350 \times 1000^{-2} \times 2000^2)$$

$$= (1300 + 1400)$$

$$= 2700 W.$$

Ans

2(a)

(i) Given that,

Primary voltage, $E_p = 220\text{ V}$

Secondary voltage, $E_s = 110\text{ V}$

(L.P.F) $100 \times 2 = 200$

Turn ratio, $\frac{N_p}{N_s} = \frac{E_p}{E_s} = \frac{220\text{ V}}{110\text{ V}} = 2$ (n)

ii) Given,

primary voltage, $V_1 = 220\text{ V}$ (E_p)

Primary current, $I_o = 0.5\text{ A}$ | Input power, $W = 30\text{ W}$

We know,

$$W = V_1 I_o \cos \phi_0$$

$$\text{or, } 30 = 220 \times 0.5 \times \cos \phi_0$$

$$\text{or, } \cos \phi_0 = \frac{30}{220 \times 0.5}$$

$$\text{or, } \phi_0 = \cos^{-1} \left(\frac{30}{220 \times 0.5} \right) = 79.1^\circ$$

Magnetising component, $I_u = I_o \sin \phi_0$

$$= 0.5 \times \sin (0.774) = 0.48\text{ A}$$

iii) Hence,

$$\phi_0 = 74.1^\circ$$

we know, $\cos \phi_0 = 0.97$

Working component $I_w = I_0 \cos \phi_0$

$$= 0.5 \times \cos (74.1)$$

$$= \frac{V_{loss}}{V_{OH}} = \frac{97}{29} = \frac{97}{29} = 0.1369 \text{ A}$$

iv) Primary loss (CW) $= I_0^2 R$

$$= (0.5)^2 \times 0.6 = 0.15 \text{ W}$$

Hence

$$R = 0.6 \Omega$$

Iron loss $= (30 - 15) \text{ W}$

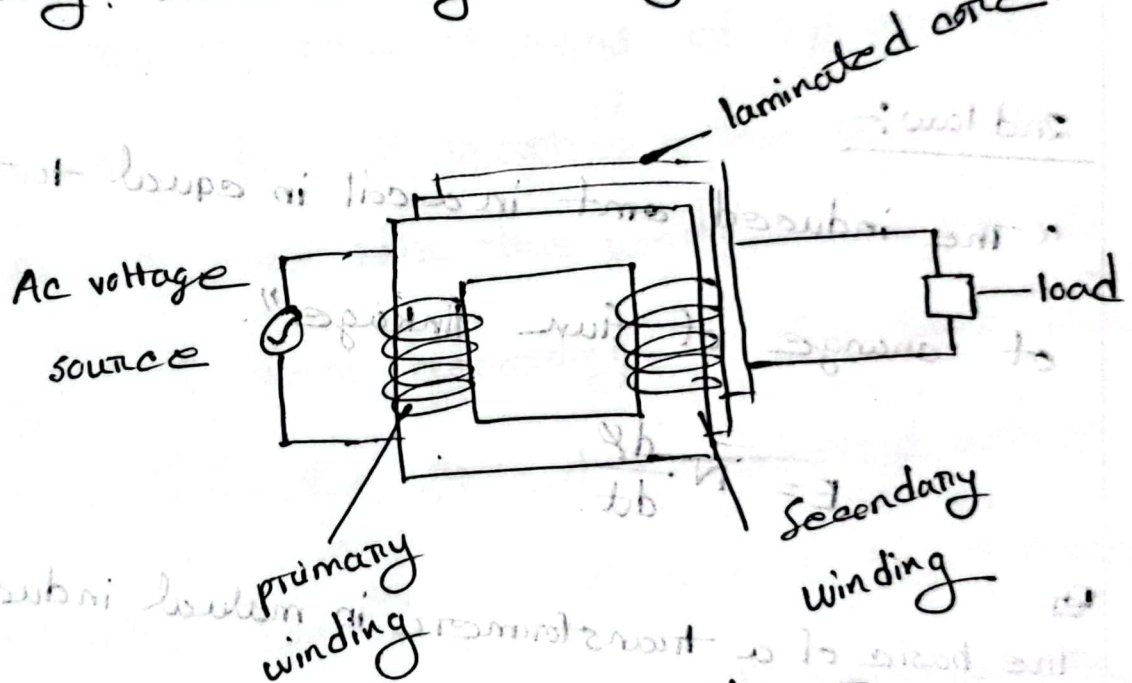
$$= 29.85 \text{ W}$$

(A)

2(A)-OR

Working Principle :-

It is essentially consist of two windings. The primary and secondary winding wound on a common laminated magnetic core. The winding connected to ac source is called primary winding. In secondary winding the core is connected.



Electrical transformer

Transformer works on the principle of Faraday's Electro-magnetic mutual Induction.

Faraday's first law :-

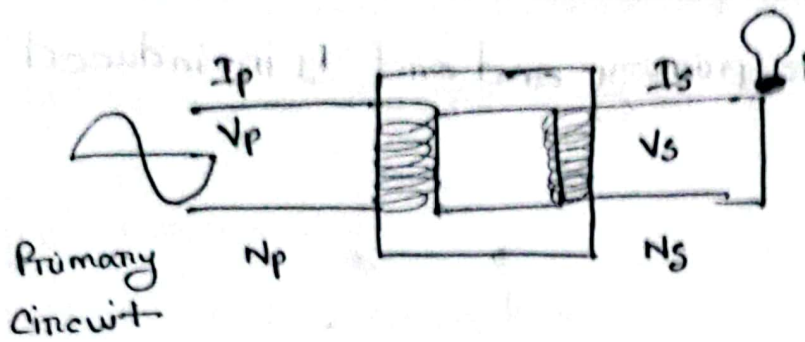
"Whenever a conductor is placed in a varying magnetic field, an electromotive force (emf/voltage) is induced. If the conductor circuit is closed, a current is induced which is called induced current."

2nd law :-

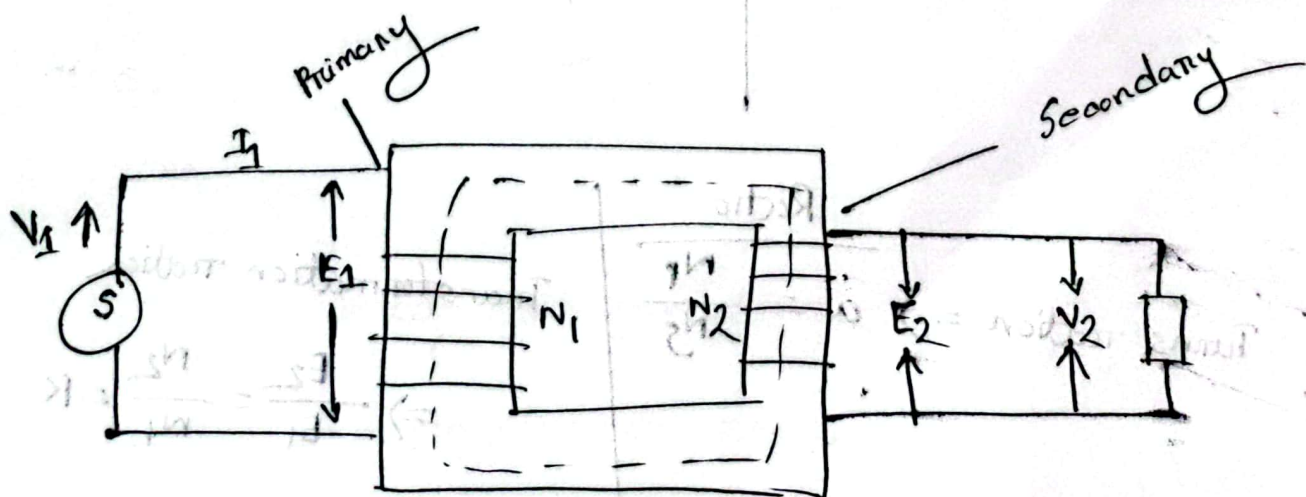
"The induced emf in a coil is equal to the rate of change of flux linkage."

$$E = N \frac{d\phi}{dt}$$

The basis of a transformer is mutual induction between two circuits linked by a common magnetic flux. If one coil of transformer is connected to a source of alternating voltage, an alternating flux is set up in the laminated core, most of the flux is linked with the other coil, according to Faraday's law of electromagnetic induction.



The alternating voltage V_1 is applied to the primary. Depending upon the number of turns of the primary (N_1) an alternating flux ϕ is set up in the core. This alternating flux ϕ links both the windings and induces emf's E_1 and E_2 in them according to Faraday's law of electromagnetic induction. If load is connected across the secondary winding, the secondary emf E_2 will cause a current I_2 to flow through the load. Consequently, terminal voltage V_2 will appear across the load.



2 (c) OR

i) shunt branch parameters from OC test (L.V. side)

$$R_0 = \frac{V^2}{P_i}$$
$$= \frac{(200)^2}{90W}$$

$$= 444\Omega$$

given,

Voltage, $V = 200V$

Power, $P_i = 90W$.

Current, $I_0 = 1.2A$.

$$I_w = \frac{V}{R_0} = \frac{200}{444} = 0.45 \text{ amp.}$$

$$I_u = \sqrt{I_0^2 - I_w^2}$$

$$= \sqrt{(1.2)^2 - (0.45)^2}$$

$$= 1.1 \text{ amp.}$$

$$X_m = \frac{V}{I_u}$$

$$= \frac{200}{1.1} = 180.2\Omega$$

3(a)

Eddy current loss: Because of time variation of flux in the magnetic material, current induced in the magnetic material. This is called eddy current. It is proportional to the square roots of the voltage.

Hysteresis loss: Hysteresis losses are

associated with the rearrangement of the magnetic domains when a magnetic material is subjected to alternating flux, power is required for the continuous reversal of molecular magnets. This power is dissipated in the form of heat and is known as "hysteresis loss".

$$\text{Hysteresis loss, } P_h = (k_h V f B_m^{1.6})$$

Working Principle:

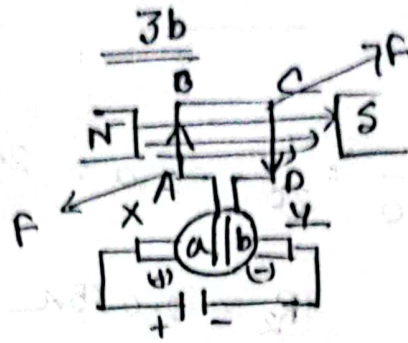


fig-1

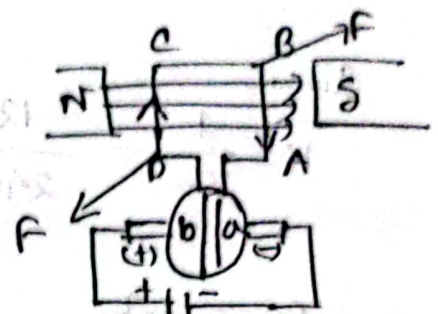


fig-2

when battery is switched on current flows through AB coil from A to B, magnetic field is from N to S. So by Fleming's left hand rule, downward force is produced on AB coil. Similarly, an upward force is produced on CD coil due to current flow from C to D. Thus the coil rotates with CD moving upward and AB moving downward [fig-1]

Hence, split ring 'a' is connected with AB coil and 'b' is connected with CD coil.

After half rotation, AB and CD coils change their position.

To change the direction of current we can use a commutator. It is consist of split rings and brush.

As the coil rotate, split ring also rotate. After half rotation coil CD attached to split ring and coil AB to a. So direction of current is D to C and B to A, and the directions reverse. [fig-2]

Now, Applying left hand rule on CD. The direction of the force downward and direction of force in AB upward mean coil rotates continuously.

This reversing of the electric current happen every half rotations and the coil continues to rotate until the battery is switched off.

3(c)

Given,

$$K = \frac{1}{a} = \frac{120}{2400} = \frac{1}{20}, \quad a = 2400/120$$

$$R_1 = 0.1 \Omega$$

$$R_2 = 0.035 \Omega$$

$$X_1 = 0.22 \Omega$$

$$X_2 = 0.012 \Omega$$

(i) High voltage side:

Here, high-voltage side is, obviously, the primary side,

Hence, value are referred to primary side

Equivalent winding resistance, $R_{01} = R_1 + R_2'$

$$= R_1 + \frac{R_2}{K^2} = (0.1) + \frac{0.035}{(1/20)^2} = 14.1 \Omega$$

$$\text{Reactance, } X_{01} = X_1 + X_2' = X_1 + \frac{X_2}{K^2} = 0.22 + \frac{0.012}{(1/20)^2}$$

$$= 5.02 \Omega$$

$$\text{Impedance, } Z_{01} = \sqrt{R_{01}^2 + X_{01}^2}$$

$$= \sqrt{(14.1)^2 + (5.02)^2}$$

$$= 15 \Omega$$

(ii) Low voltage side.

Equivalent winding resistance, $R_{02} = R_1' + R_2$

$$= K^2 R_1 + R_2 = \left(\frac{1}{20}\right)^2 \times 0.1 + 0.035$$

$$= 0.03525 \Omega$$

Reactance, $X_{02} = X_1' + X_2$

$$= X_1 K^2 + X_2$$

$$= \left(\frac{1}{20}\right)^2 \times 0.22 + 0.012$$

$$= 0.01255 \Omega$$

Impedance, $Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$

$$= \sqrt{(0.03525)^2 + (0.01255)^2}$$

$$= 0.0374 \Omega$$